From Prediction to Shared Decision Making - A Patient Preference-Centered Treatment Recommendation

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Clinical Prediction Alone is Not Enough

ML Limitations

ML/AI models can estimate treatment effects very well, but can't make the final call

Context

Each patient is unique beyond just their outward characteristics

Physicians must balance both the physical realities/needs of a patient along with their emotional and preferential response to treatment options

Problem statement

Properly structuring patient needs can be a difficult task, often leading to patients preferences being disregarded or ignored

Dataset B

- 1,000 Future Patients
- 5 tokenized weights (WP1 to WP5) based upon a 100 point distribution system
- Each weight corresponds to one outcome used in the CDS from the last challenge
- Token totals do not always sum to 100 due potentially to rounding, survey interface
- Preference weights are normalized to preserve relative importance for each outcome
- Intentionally excludes the X,Y, and T variables from previous dataset, meaning this is explicitly based on a patient's personal preferences
- <u>Note:</u> next slide references provided scaling reasoning

Scaling Appropriately

- Provided monetary anchors were used for w1,2, and 3; but this excludes two of the 5 output scales
- The last two (w4 and w5) were standardized as penalty terms to maintain comparability
- Rehab time (w4) was converted to a penalty of $1,000 per week of recovery - ends up being a very heavy penalty on average so this is adjusted to what's typical in the data, then capped out at $25k
- Out-of-pocket cost (w5) was multiplied by 1,000 then treated as a direct $ penalty, since it is already in dollars at this point it directly matches the other weights at this point

Methodology

Preference-Weighted Decision Framework

We convert treatment effects into a common value scale, weigh them based on preferences, and calculate a final score

1) Utilize the treatment effects from DataSet A

2) Translate these outcomes to a standard scale using the willingness-to-pay anchors provided (in this case $)

3) Normalize the patient preference weights from Dataset B

4) Gather values into a clear decision score (if score is above 0, recommend surgery)

5) Recommend treatment to assist in the clinical assessment and choice

$w_k = WP_k / \sum WP_j$ For j = 1 to 5, k being different outcomes

Wk - Patient Preference Weight, sum to 1ish

WPk - Token Score (1 to 5)

V_k = willingness to pay $ (ak) * direction(dk) * TEk (from Dataset A)

Vk - Value difference between surgery and conservative care, ak is the provided $ amount, dk +1 for better and -1 for worse

Dk- +1 if adds value, -1 if the outcome takes value

$Score = \sum w_k \times V_k$ for k= 1 to 5

Mathematical Example for 117 (For reference)

WP1 = 32.84230393

WP2 = 11.27050540

WP3 = 59.58044664

WP4 = 18.51752513

WP5 = 25.09631866

TE1	0.250474
TE2	-0.199238
TE3	0.250286
TE4	19980.826807
TE5	5.994707

From DataSet A

$V1=+25{,}000\times0.250474=$ **+$6,261.85**

$V2=-15{,}000\times(-0.199238)=$ **+$2,988.57**

$V3=+25{,}000\times0.250286=$ **+$6,257.15**

$V4=-\tilde{TE}4\cdot25000\approx-0.498979\cdot25000=$ **−$12474.48** (**Normalized rehab** effect)

$V5=-5.994707\times1{,}000=$ **−$5,994.71**

Normalized rehab: 1st percentile - avg TE4/99th percentile-avg TE4

w1 = 0.222951263

w2 = 0.076510266

w3 = 0.404464189

w4 = 0.125706943

w5 = 0.170367339

Score117$=w1V1+w2V2+w3V3+w4V4+w5V5$

Score117$\approx1396.09+228.66+2530.79-1568.13-1021.30$

Score117 = 1566.11>0, Surgery

Example

Patient 117 - John

Preferences: Higher weight on long-term function (w3 of 0.44) and long-term pain reduction (w1 = 0.23). Much lower concern on rehab time (w4 of 0.12) and cost (w5 of 0.17)

Trade-Offs (Derived from Dataset A): Surgery provides a +25% long-term pain reduction, +25% functional improvement, -20% short term pain reduction,$6,000 higher out-of-pocket cost, and a much longer rehab time

Recommendation: Surgery

Considering this patient’s weighted preferences and the clinical evidence provided from the previous model, we can reasonably suggest surgery as the best option for this patient

CDS Interface

Clinical Decision Support Tool
MSK Treatment Recommendation System

Patient ID: 00003

RECOMMENDED TREATMENT
Surgery
Confidence: Strong | Preference-Weighted Score: +0.68

Treatment Outcome Comparison

Outcome Domain	Conservative Care	Surgery
Long-term Pain Relief Weight: 35%	0%	+39.5%
Treatment Pain Weight: 10%	0%	-34.1%
Long-term Function Weight: 30%	0%	+30.0%
Treatment Cost Weight: 15%	$0	$24,666
Recovery Time Weight: 10%	0 days	7.1 days

Key Considerations
- Higher upfront cost but strong long-term gains
- Moderate increase in treatment discomfort
- Patient priority: Long-term pain & function (65%)

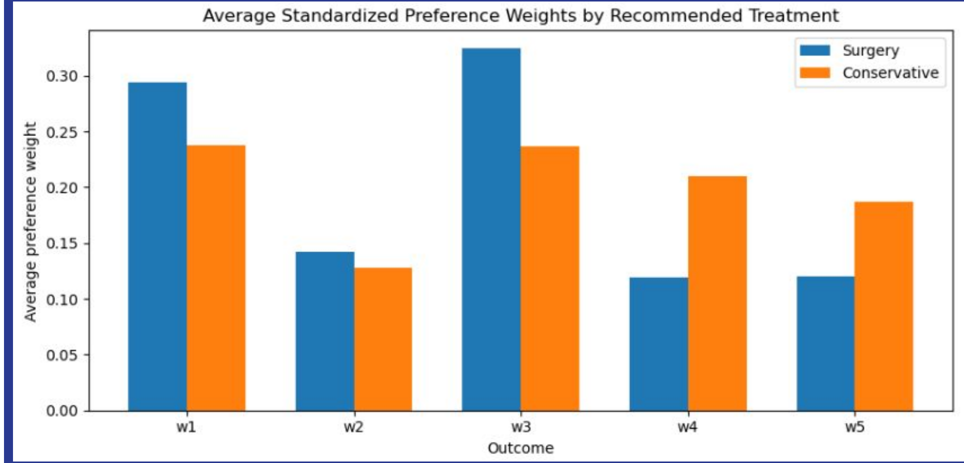
Patient Preference Distribution

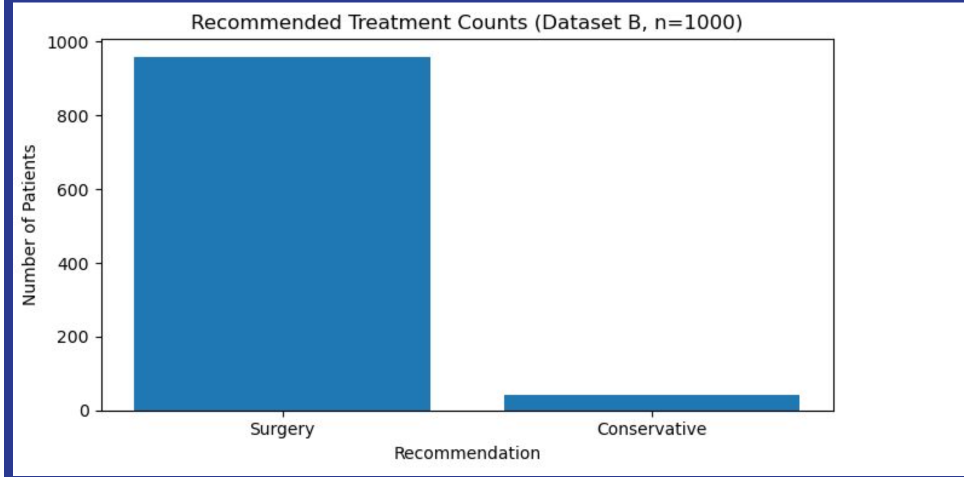

How clinicians can use the CDS:

1. Patient completes a 100 - point preference survey before their evaluation
2. Clinician reviews the CDS and views the patient’s records
3. The CDS provides treatment recommendations based on historical clinical evidence as well as the patient’s preferences
4. Clinician reviews these outcomes with the patient, discussing benefits and trade-offs, then expressing their own expert insight
5. The healthcare professional can then further consult the patient with their personal values and priorities
6. The clinician and patient arrive at a shared decision on what treatment option will be best

Results

- Patients recommended surgery often place greater weight on
    - Long-term functional improvement (w3),. This is by far the most influential preference, patients wanting long term function are likely to be recommended surgery
    - Long-term pain reduction (w1 slightly)
- Patients recommended conservative care place greater weight on:
    - Rehabilitation burden (w4)
    - Out-of-pocket cost (w5)
- Treatment related pain (w2) matters, but this along with w1 are not the primary differentiators when considering long-term payoffs





Conclusions

- Treatment decisions involve noteworthy trade offs across different patient preferences
- Clinical evidence is great, but it isn’t everything when it comes to healthcare, especially considering patient values
  - The same clinical data in two patients may recommend an ideal situation to one patient, and a very inconvenient one to another
- This CDS framework combines
  - Treatment effects based on clinical evidence
  - Patient provided preferences in their outcomes
- The best decision is not just the most accurate, but is also the best fit for the clinician and the patient